

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)
ON
AI FRAUD DETECTION IN CRYPTOCURRENCY USING MACHINE LEARNING
submitted in partial fulfilment of the requirements for the award of degree of
BACHELOR OF ENGINEERING
In
COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**AI FRAUD DETECTION IN CRYPTOCURRENCY USING MACHINE LEARNING**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Shikha Tuteja**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Date : 6 May 2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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1. Introduction

Cryptocurrency has become one of the fastest-growing digital financial technologies in the modern world. Blockchain systems enable decentralized and transparent transactions without depending on traditional banking institutions.

However, the rapid growth of cryptocurrency platforms has also increased fraudulent activities such as phishing attacks, fake investment schemes, wallet theft, Ponzi schemes, rug-pull scams, and money laundering.

Traditional fraud detection systems mainly depend on predefined rules and manual monitoring techniques.

These approaches often fail to identify newly emerging fraud strategies because cybercriminals continuously change their transaction behaviour. As blockchain transaction volume increases, manual monitoring becomes inefficient and difficult to manage.

1.1 Background of Cryptocurrency Fraud

Cryptocurrency has rapidly transformed the digital financial sector by enabling decentralized and transparent online transactions. Blockchain technology allows users to transfer digital assets without depending on traditional banking institutions or centralized authorities. Due to its global accessibility and fast transaction capability, cryptocurrency adoption has increased significantly in recent years.

However, the rapid expansion of cryptocurrency platforms has also increased cybercrime and fraudulent activities within blockchain ecosystems. Fraudsters exploit the anonymous and decentralized nature of blockchain systems to perform illegal activities such as phishing attacks, fake investment schemes, Ponzi scams, wallet theft, and money laundering.

Traditional fraud detection systems are often unable to handle modern blockchain fraud because transaction behaviour changes continuously. This creates a need for intelligent and automated fraud detection systems capable of analysing large-scale blockchain transaction data efficiently.

1.2 Role of Machine Learning in Fraud Detection

Machine learning has emerged as an effective solution for identifying fraudulent activities within cryptocurrency systems. Unlike traditional rule-based security methods, machine learning models can learn transaction behaviour patterns from historical data and automatically identify suspicious activities.

Machine learning algorithms analyse features such as transaction amount, wallet connectivity, transaction timing, transfer frequency, and behavioural anomalies to classify transactions as legitimate or fraudulent.

The use of machine learning provides several advantages including:

- Automated fraud detection
- Real-time transaction monitoring

2. METHODOLOGY

The proposed cryptocurrency fraud detection system follows a structured machine learning workflow designed for analysing blockchain transaction behaviour and identifying suspicious activities automatically.

The methodology begins with data collection from publicly available cryptocurrency transaction datasets and blockchain repositories. After collecting the data, preprocessing operations such as missing value handling, duplicate removal, feature normalization, and scaling are applied to improve dataset quality and model stability.

Feature engineering techniques are then used to extract meaningful behavioural indicators such as transaction frequency, wallet age, transfer intervals, transaction amount, and network connectivity patterns. These features help machine learning models understand the difference between legitimate and fraudulent transaction behaviour.

Multiple supervised learning algorithms including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost are trained independently using the processed datasets. The datasets are divided using an 80/20 train-test split strategy for proper model validation and performance evaluation.

Finally, evaluation metrics such as Accuracy, Precision, Recall, F1-Score, and AUC-ROC are used to compare model performance and identify the most effective fraud detection algorithm.

3. TOOLS AND TECHNOLOGIES

The project was implemented using Python programming language along with several machine learning and data analysis libraries.

The major tools and technologies used in this project include:

- Python for machine learning model development
- Pandas for data preprocessing and dataset handling
- NumPy for numerical computations and matrix operations
- Scikit-learn for machine learning algorithms and evaluation metrics
- XGBoost for boosting-based fraud classification
- Matplotlib for graph generation and result visualization
- Visual Studio Code as the primary development environment
- GitHub for version control and project repository management

The datasets used in this project were collected from Kaggle cryptocurrency repositories and blockchain transaction datasets.

4. IMPLEMENTATION

The implementation phase focused on developing a machine learning-based cryptocurrency fraud detection framework capable of analysing blockchain transaction behaviour and identifying suspicious activities.

The datasets were imported into the Python environment using Pandas libraries. Data preprocessing techniques such as cleaning, normalization, duplicate removal, and feature scaling were applied to improve dataset consistency and quality.

Artificial fraud labels were generated using abnormal transaction behaviour patterns to train the machine learning models effectively. The datasets were divided into training and testing partitions using an 80/20 split strategy.

Machine learning models including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost were implemented independently using Scikit-learn and XGBoost libraries.

After training, evaluation metrics such as Accuracy, Precision, Recall, F1-Score, and AUC-ROC were calculated for each model. Comparative graphs were generated to analyse overall model performance visually. Because apparently no project is considered real until there are colourful graphs proving the computer suffered for science.

5. MAJOR FINDINGS / OUTCOMES / RESULTS

The experimental analysis demonstrated that machine learning algorithms can effectively identify fraudulent cryptocurrency transactions using behavioural transaction analysis.

Among all evaluated models, XGBoost achieved the highest overall fraud detection performance because of its ability to capture complex behavioural patterns and hidden transaction relationships.

Random Forest also demonstrated strong performance, especially for noisy blockchain transaction datasets. Logistic Regression and Decision Tree achieved moderate results, while Support Vector Machine showed lower performance on the selected dataset.

The observed performance ordering was:

XGBoost > Random Forest > Decision Tree > Logistic Regression > Support Vector Machine

The results also highlighted the importance of Recall in fraud detection systems because false negatives may result in financial losses, asset theft, and reduced user trust.

The generated graphs and evaluation metrics confirmed that ensemble learning methods are highly effective for cryptocurrency fraud detection applications.

6. CONCLUSION AND FUTURE SCOPE

This project successfully developed a machine learning-based cryptocurrency fraud detection framework using blockchain transaction analysis and behavioural pattern recognition.

Multiple supervised learning algorithms including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost were implemented and evaluated under standardized experimental conditions.

Experimental analysis showed that ensemble learning models such as XGBoost and Random Forest achieved superior fraud detection performance compared to traditional classification approaches because of their ability to handle complex transaction relationships effectively.

The project also highlighted important challenges associated with cryptocurrency fraud detection including scalability limitations, privacy concerns, biased datasets, and ethical AI deployment issues.

Future improvements may include:

- Deep learning integration
- Graph Neural Networks for blockchain analytics
- Real-time blockchain monitoring systems
- Cross-chain fraud detection
- AI-driven adaptive security frameworks

The combination of machine learning and blockchain analytics provides a promising direction for improving digital financial security systems and reducing fraudulent activities within cryptocurrency ecosystems.

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APPENDICES

Appendix A

Contains generated graphs for Accuracy, Precision, Recall, F1-Score, and AUC-ROC evaluation metrics.

Appendix B

Includes screenshots of implementation output, model training results, and comparative analysis graphs.

Appendix C

Contains dataset information, software tools, and hardware requirements used during project implementation.